Fall 2025 CPTS530 Homework 4
MATH 448-548
Numerical Analysis

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Contents

Problem 1

Problem Statement (Textbook 4.1.7):

Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $(B - I)$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

Solution:

Problem 2

Problem Statement (Textbook 4.1.12):

Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

Solution:

Problem 3

Problem Statement (Textbook 4.1.17):

A square matrix A is said to be **skew-symmetric** if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x.

Solution:

Problem 4

Problem Statement (Textbook 4.1.18):

(Continuation) Prove that the diagonal elements of a skew-symmetric matrix are **0**. Also, prove that the determinant is 0 when the matrix is of odd order.

Solution:

Problem 5

Problem Statement (Textbook 4.1.19):

(Continuation) Let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x, $x^T A x = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

Solution:

Problem 6

Part (a)

Problem Statement (Textbook 4.2.1a):

Prove these facts, needed in the proof of Theorem 2.

(a) If U is upper triangular and invertible, then U^{-1} is upper triangular.

Solution:

Part (b)

Problem Statement (Textbook 4.2.1b):

Prove these facts, needed in the proof of Theorem 2.

(b) The inverse of a unit lower triangular matrix is unit lower triangular.

Solution:

Part (c)

Problem Statement (Textbook 4.2.1c):

Prove these facts, needed in the proof of Theorem 2.

(c) The product of two upper (lower) triangular matrices is upper (lower) triangular.

Solution:

Problem 7

Problem Statement (Textbook 4.2.5):

Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.

Solution:

Appendix